Université Catholique d'Afrique Centrale (UCAC)

Institut Catholique de Yaoundé (ICY)

COURSE TITLE: OPERATION RESEARCH
DEPARTMENTS: BEM3
LECTURER: Dr. ARREY Marinus
INSTRUCTIONS: ANSWER ALL QUESTIONS

FIRST SEMESTER EXAMINATION

Question 1: (8 marks)

A company manufactures three products namely X, Y and Z. Each product requires processing on three machines turning, milling and grinding. Product X requires 10 hours of turning, 5 hours of milling and 1 hour of grinding. Product Y requires 5 hours of turning, 4 hours of milling and 2 hours of grinding. In the coming planning period, 2700 hours of turning, 2200 hours of milling and 500 hours of grinding are available. The profit contribution of X, Y and Z are 10 frs, 15 frs and 20 frs per unit respectively. Find the optimal product mix to maximize the profit.

Question 2: (4 marks)

Find the transportation cost for the following by using the list cost method.

	A	B	C	D	E	F	G	Supply
A	5	6	4	3	7	5	4	7000
B	9	4	3	4	3	2	1	4000
C	8	4	2	5	4	8	3	10000
Demand	1500	2000	4500	4000	2500	3500	3000	

Question 3: (8 marks)

$$\text{Minimize } Z = 6X_1 + 3X_2$$

$$\text{St} \quad 2X_1 + 4X_2 \geq 16$$

$$4X_1 + 3X_2 \geq 24$$

$$X_1, X_2 \geq 0$$

a) Determine the dual of the following LPP.

b) Determine the optimal solution of the primal as well as that of the dual using the simplex method.

Good Luck!

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